Takumi **Shinohara**, Ph.D.

Postdoctoral Researcher
Division of Decision and Control Systems,
KTH Royal Institute of Technology
Malvinas Väg 10, 114 28, Stockholm, Sweden

tashin@kth.se | Takumi Shinohara | Takumi Shinohara

Employment

May 2025 – present	**Postdoctoral Researcher, KTH Royal Institute of Technology**, Sweden *Research on learning and control of networked controlled systems with Prof. Karl Henrik Johansson.*
Oct. 2024 – Mar. 2025	**Visiting Researcher, Keio University**, Japan *Research on secure control systems with Prof. Toru Namerikawa.*
Apr. 2018 – Apr. 2025	**Consultant, Mitsubishi Research Institute, Inc.**, Japan *Research, study, and consult on the cybersecurity policy for the Japanese Government (e.g., METI, MIC, and NISC) and consult private companies with cybersecurity issues.*

Education

Sep. 2021 – Sep. 2024	**Ph.D. in Engineering** of Keio University, Japan Thesis Title: *Secure state estimation under sensor attacks* Advisor: Prof. Toru Namerikawa
Apr. 2016 – Mar. 2018	**Master in Engineering** of Keio University, Japan Thesis Title: *Zero-stealthy attacks in cyber-physical systems and secure state estimation in adversarial environments* Advisor: Prof. Toru Namerikawa
Apr. 2012 – Mar. 2016	**Bachelor in Engineering** of Keio University, Japan Thesis Title: *SLAM problem for UAV with considering computational load and unordinary observations* Advisor: Prof. Toru Namerikawa

List of Publications

I. Journal Articles

[J1] Takumi Shinohara and Toru Namerikawa, "Relationship between the number of agents and sparse observability index," *IEEE Open Journal of Control Systems*, vol. 4, pp. 144–155, 2025.

[J2] Takumi Shinohara and Toru Namerikawa, "Optimal security investment problem for secure state estimation on cyber-physical systems," *IEEE Transactions on Automatic Control*, vol. 70, no. 2, pp. 1244–1251, 2025.

[J3] Takumi Shinohara and Toru Namerikawa, "Optimal resilient sensor placement problem for secure state estimation," *Automatica*, vol. 160, 111454, 2024.

[J4] Takumi Shinohara and Toru Namerikawa, "Distributed secure state estimation with a priori sparsity information," *IET Control Theory & Applications*, vol. 16, no. 11, pp. 1086−1097, 2022.

[J5] Takumi Shinohara, Toru Namerikawa, and Zhihua Qu, "Resilient reinforcement in secure state estimation against sensor attacks with *a priori* information," *IEEE Transactions on Automatic Control*, vol. 64, no. 12, pp. 5024−5038, 2019.

[J6] Takumi Shinohara and Toru Namerikawa, "Reach set-based secure state estimation against sensor attacks with interval hull approximation," *SICE Journal of Control, Measurement, and System Integration*, vol. 11, no. 5, pp. 399−408, 2018.

[J7] Takumi Shinohara and Toru Namerikawa, "Perfect stealthy attacks in cyber-physical systems," *Transactions of the Society of Instrument and Control Engineers*, vol. 54, no. 3, pp. 309−319, 2018. (in Japanese)

[J8] Takumi Shinohara and Toru Namerikawa, "On the vulnerabilities due to manipulative zero-stealthy attacks in cyber-physical systems," *SICE Journal of Control, Measurement, and System Integration*, vol. 10, no. 6, pp. 563−570, 2017.

[J9] Takashi Irita, Takumi Shinohara and Toru Namerikawa, "Detection of replay attack on smart grid with code signal and bargaining game," *Transactions of the Society of Instrument and Control Engineers*, vol. 52, no. 9, pp. 498−506, 2016. (in Japanese)

II. Referred Conference Papers

[C1] Takumi Shinohara and Toru Namerikawa, "Secure state estimation, attack detection, and safe control against sensor attacks with side initial state information," in *Proc. 64th IEEE Conference on Decision and Control*. (accepted for presentation)

[C2] Takumi Shinohara and Toru Namerikawa, "Security measure implementation for distributed state estimation," in *Proc. 5th IFAC Workshop on Cyber-Physical Human Systems,* Antalya, Türkiye, 2024.

[C3] Takumi Shinohara and Toru Namerikawa, "Secure state estimation for multi-agent systems: On the relationship between the number of agents and system resilience," in *Proc. 2023 American Control Conference*, San Diego, CA, 2023, pp. 1006−1011.

[C4] Takumi Shinohara and Toru Namerikawa, "Reach set-based attack resilient state estimation against

omniscient adversaries," in *Proc. 2018 American Control Conference*, Milwaukee, WI, 2018, pp. 5813−5818.

[C5] Takumi Shinohara and Toru Namerikawa, "Manipulative zero-stealthy attacks in cyber-physical systems: Existence space of feasible attack objectives," in *Proc. 1st IEEE Conference on Control Technology and Applications*, Kohala Coast, HI, 2017, pp. 1123−1128.

[C6] Takumi Shinohara and Toru Namerikawa, "SLAM for a small UAV with compensation for unordinary observations and convergence analysis," in *Proc. 2016 55th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsukuba, Japan, 2016, pp. 1252−1257.

Honors and Awards

- Wallenberg Postdoctoral Scholarship, Knut and Alice Wallenberg Foundation, 2025–present.
- Fujiwara award (Best Ph.D. student award), Keio University, 2025.
- Best presentation award, 65th Japan Joint Automatic Control Conference, 2022.
- Best presentation award, 59th Japan Joint Automatic Control Conference, 2016.

Review Activities

- *Journals*
  - IEEE Transactions on Automatic Control
  - Automatica
  - IEEE Transactions on Control of Network Systems
  - IEEE Transactions on Cybernetics
  - IEEE Control Systems Letters
  - IEEE Journal on Selected Areas in Communications
- *Conferences*
  - IEEE Conference on Decision and Control
  - American Control Conference
  - IEEE Vehicular Technology Conference